

Chapter 3

Aircraft Construction

Introduction

An aircraft is a device that is used, or intended to be used, for flight according to the current Title 14 of the Code of Federal Regulations (14 CFR) part 1, Definitions and Abbreviations. Categories of aircraft for certification of airmen include airplane, rotorcraft, glider, lighter-than-air, powered-lift, powered parachute, and weight-shift control aircraft. Title 14 CFR part 1 also defines airplane as an engine-driven, fixed-wing aircraft that is supported in flight by the dynamic reaction of air against its wings. Another term, not yet codified in 14 CFR part 1, is advanced avionics aircraft, which refers to an aircraft that contains a global positioning system (GPS) navigation system with a moving map display, in conjunction with another system, such as an autopilot. This chapter provides a brief introduction to the structure of aircraft and uses an airplane for most illustrations. Light Sport Aircraft (LSA), such as weight-shift control aircraft, balloon, glider, powered parachute, and gyroplane, have their own handbooks to include detailed information regarding aerodynamics and control.

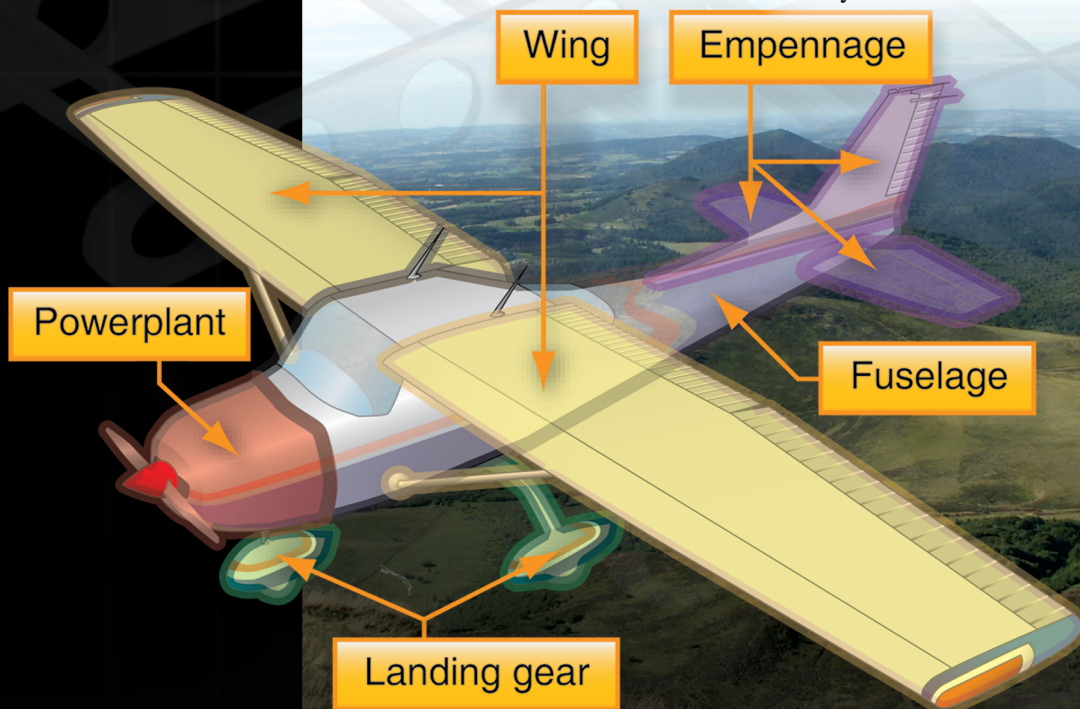


Figure 1-1. Airplane (14 CFR)

Aircraft Design, Certification, and Airworthiness

The FAA certifies three types of aviation products: aircraft, aircraft engines, and propellers. Each of these products has been designed to a set of airworthiness standards. These standards are parts of Title 14 of the Code of Federal Regulations (14 CFR), published by the FAA. The airworthiness standards were developed to help ensure that aviation products are designed with no unsafe features. Different airworthiness standards apply to the different categories of aviation products as follows:

- Normal, Utility, Acrobatic, and Commuter Category Airplanes— 14 CFR part 23
- Transport Category Airplanes—14 CFR part 25
- Normal Category—14 CFR part 27
- Transport Category Rotorcraft—14 CFR part 29
- Manned Free Balloons—14 CFR part 31
- Aircraft Engines—14 CFR part 33
- Propellers—14 CFR part 35

Some aircraft are considered “special classes” of aircraft and do not have their own airworthiness standards, such as gliders and powered lift. The airworthiness standards used for these aircraft are a combination of requirements in 14 CFR parts 23, 25, 27, and 29 that the FAA and the designer have agreed are appropriate for the proposed aircraft.

The FAA issues a Type Certificate (TC) for the product when they are satisfied it complies with the applicable airworthiness standards. When the TC is issued, a Type Certificate Data Sheet (TCDS) is generated that specifies the important design and operational characteristics of the aircraft, aircraft engine, or propeller. The TCDS defines the product and are available to the public from the FAA website at www.faa.gov.

A Note About Light Sport Aircraft

Light sport aircraft are not designed according to FAA airworthiness standards. Instead, they are designed to a consensus of standards agreed upon in the aviation industry. The FAA has agreed the consensus of standards is acceptable as the design criteria for these aircraft. Light sport aircraft do not necessarily have individually type certificated engines and propellers. Instead, a TC is issued to the aircraft as a whole. It includes the airframe, engine, and propeller.

Aircraft, aircraft engines, and propellers can be manufactured one at a time from the design drawings, or through an FAA approved manufacturing process, depending on the size and capabilities of the manufacturer. During the manufacturing

process, each part is inspected to ensure that it has been built exactly according to the approved design. This inspection is called a conformity inspection.

When the aircraft is complete, with the airframe, engine, and propeller, it is inspected and the FAA issues an airworthiness certificate for the aircraft. Having an airworthiness certificate means the complete aircraft meets the design and manufacturing standards, and is in a condition for safe flight. This airworthiness certificate must be carried in the aircraft during all flight operations. The airworthiness certificate remains valid as long as the required maintenance and inspections are kept up to date for the aircraft.

Airworthiness certificates are classified as either “Standard” or “Special.” Standard airworthiness certificates are white, and are issued for normal, utility, acrobatic, commuter, or transport category aircraft. They are also issued for manned free balloons and aircraft designated as “Special Class.”

Special airworthiness certificates are pink, and are issued for primary, restricted, and limited category aircraft, and light sport aircraft. They are also issued as provisional airworthiness certificates, special flight permits (ferry permits), and for experimental aircraft.

More information on airworthiness certificates can be found in Chapter 9, in 14 CFR parts 175-225, and also on the FAA website at www.faa.gov.

Lift and Basic Aerodynamics

In order to understand the operation of the major components and subcomponents of an aircraft, it is important to understand basic aerodynamic concepts. This chapter briefly introduces aerodynamics; a more detailed explanation can be found in Chapter 5, Aerodynamics of Flight.

Four forces act upon an aircraft in relation to straight-and-level, unaccelerated flight. These forces are thrust, lift, weight, and drag. *[Figure 3-1]*

Thrust is the forward force produced by the powerplant/propeller. It opposes or overcomes the force of drag. As a general rule, it is said to act parallel to the longitudinal axis. This is not always the case as explained later.

Drag is a rearward, retarding force and is caused by disruption of airflow by the wing, fuselage, and other protruding objects. Drag opposes thrust and acts rearward parallel to the relative wind.

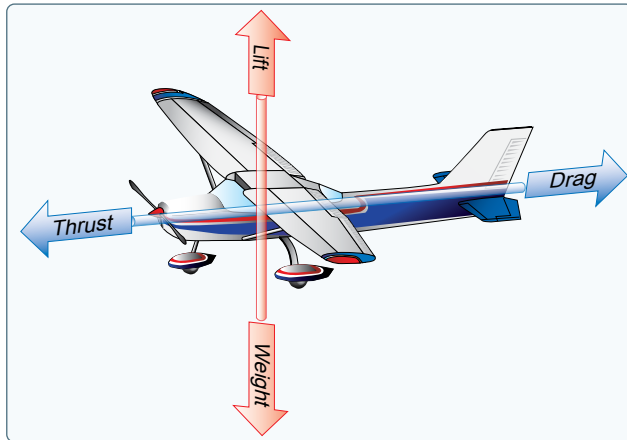


Figure 3-1. *The four forces.*

Weight is the combined load of the aircraft itself, the crew, the fuel, and the cargo or baggage. Weight pulls the aircraft downward because of the force of gravity. It opposes lift and acts vertically downward through the aircraft's center of gravity (CG).

Lift opposes the downward force of weight, is produced by the dynamic effect of the air acting on the wing, and acts perpendicular to the flight path through the wing's center of lift (CL).

An aircraft moves in three dimensions and is controlled by moving it about one or more of its axes. The longitudinal, or roll, axis extends through the aircraft from nose to tail, with the line passing through the CG. The lateral or pitch axis extends across the aircraft on a line through the wing tips, again passing through the CG. The vertical, or yaw, axis passes through the aircraft vertically, intersecting the CG. All control movements cause the aircraft to move around one or more of these axes and allows for the control of the aircraft in flight. [Figure 3-2]

One of the most significant components of aircraft design is CG. It is the specific point where the mass or weight of an aircraft may be said to center; that is, a point around which, if the aircraft could be suspended or balanced, the aircraft would remain relatively level. The position of the CG of an aircraft determines the stability of the aircraft in flight. As the CG moves rearward (towards the tail), the aircraft becomes more and more dynamically unstable. In aircraft with fuel tanks situated in front of the CG, it is important that the CG is set with the fuel tank empty. Otherwise, as the fuel is used, the aircraft becomes unstable. [Figure 3-3] The CG is computed during initial design and construction and is further affected by the installation of onboard equipment, aircraft loading, and other factors.

Major Components

Although airplanes are designed for a variety of purposes, most of them have the same major components. [Figure 3-4] The overall characteristics are largely determined by the original design objectives. Most airplane structures include a fuselage, wings, an empennage, landing gear, and a powerplant.

Fuselage

The fuselage is the central body of an airplane and is designed to accommodate the crew, passengers, and cargo. It also provides the structural connection for the wings and tail assembly. Older types of aircraft design utilized an open truss structure constructed of wood, steel, or aluminum tubing. [Figure 3-5] The most popular types of fuselage structures used in today's aircraft are the monocoque (French for "single shell") and semimonocoque. These structure types are discussed in more detail under aircraft construction later in the chapter.

Wings

The wings are airfoils attached to each side of the fuselage and are the main lifting surfaces that support the airplane in

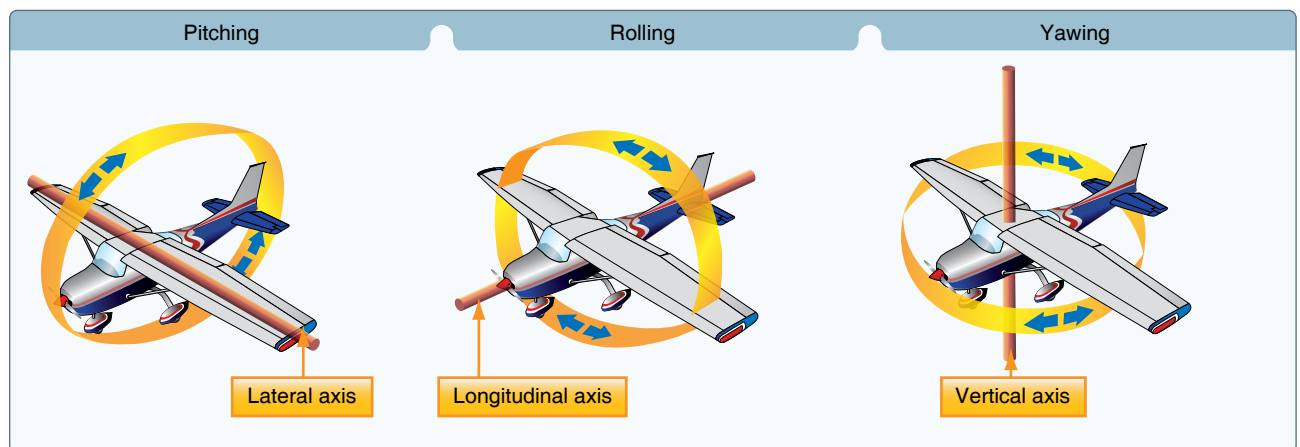


Figure 3-2. *Illustrates the pitch, roll, and yaw motion of the aircraft along the lateral, longitudinal, and vertical axes, respectively.*

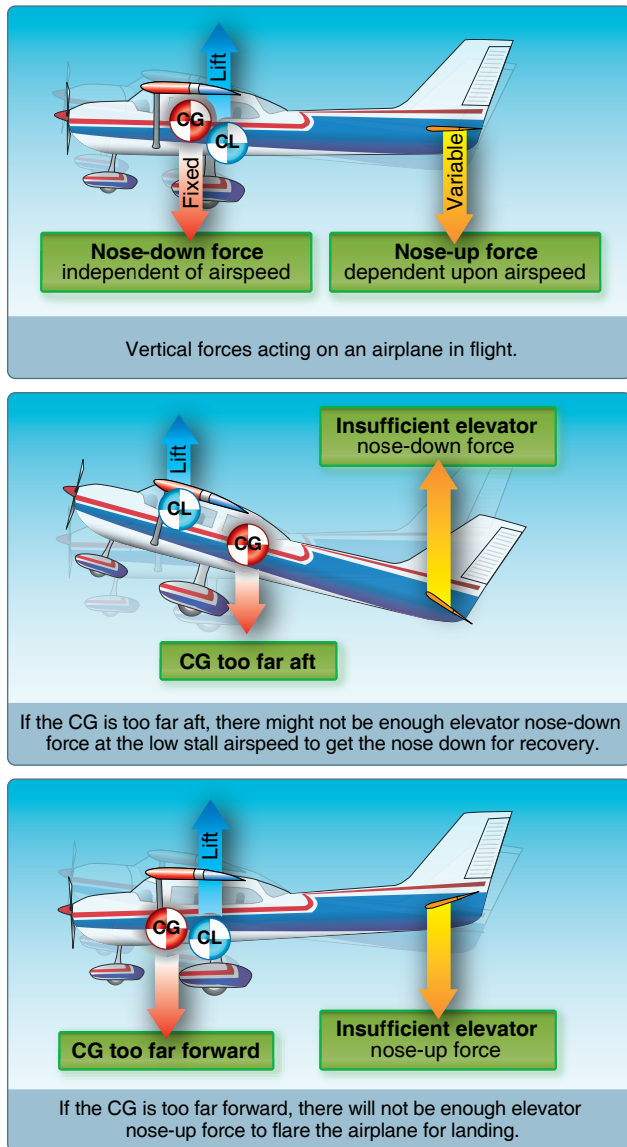


Figure 3-3. Center of gravity (CG).

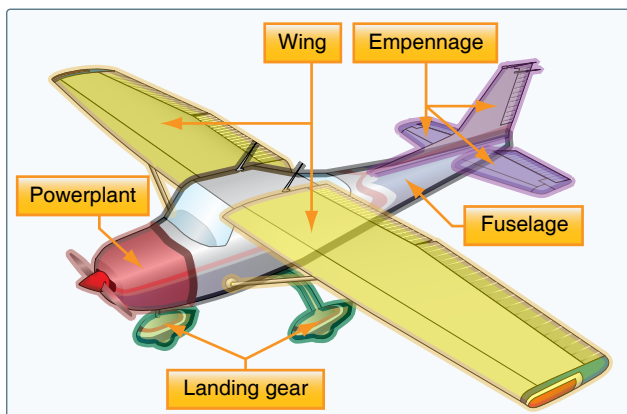


Figure 3-4. Airplane components.

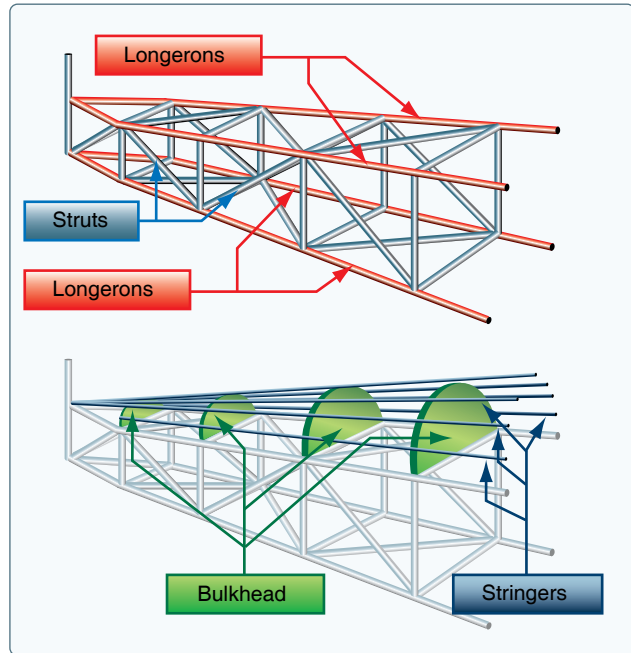


Figure 3-5. Truss-type fuselage structure.

flight. There are numerous wing designs, sizes, and shapes used by the various manufacturers. Each fulfills a certain need with respect to the expected performance for the particular airplane. How the wing produces lift is explained in Chapter 5, Aerodynamics of Flight.

Wings may be attached at the top, middle, or lower portion of the fuselage. These designs are referred to as high-, mid-, and low-wing, respectively. The number of wings can also vary. Airplanes with a single set of wings are referred to as monoplanes, while those with two sets are called biplanes. [Figure 3-6]

Many high-wing airplanes have external braces, or wing struts that transmit the flight and landing loads through the struts to the main fuselage structure. Since the wing struts are usually attached approximately halfway out on the wing, this type of wing structure is called semi-cantilever. A few high-wing and most low-wing airplanes have a full cantilever wing designed to carry the loads without external struts.

The principal structural parts of the wing are spars, ribs, and stringers. [Figure 3-7] These are reinforced by trusses, I-beams, tubing, or other devices, including the skin. The wing ribs determine the shape and thickness of the wing (airfoil). In most modern airplanes, the fuel tanks are either an integral part of the wing's structure or consist of flexible containers mounted inside of the wing.

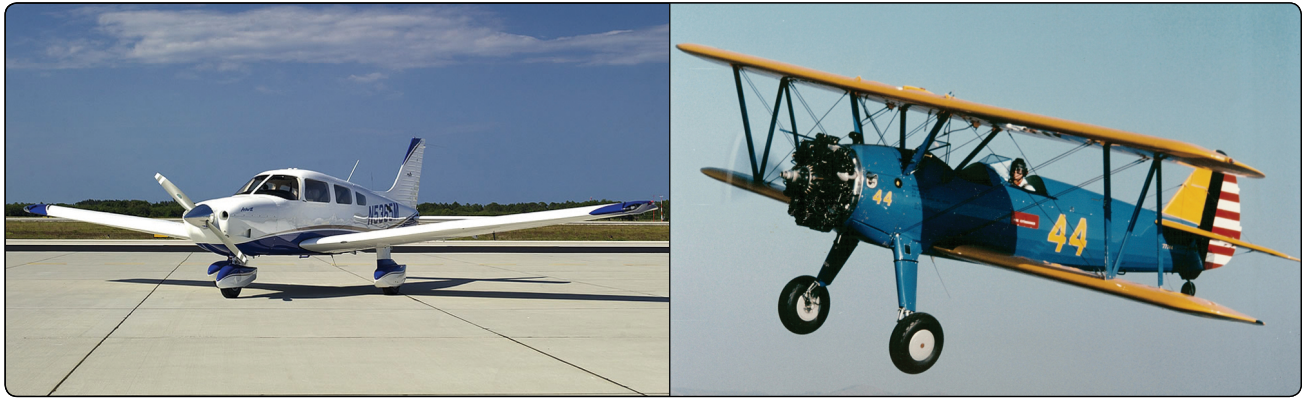


Figure 3-6. Monoplane (left) and biplane (right).

Attached to the rear, or trailing edges, of the wings are two types of control surfaces referred to as ailerons and flaps. Ailerons extend from about the midpoint of each wing outward toward the tip, and move in opposite directions to create aerodynamic forces that cause the airplane to roll. Flaps extend outward from the fuselage to near the midpoint of each wing. The flaps are normally flush with the wing's surface during cruising flight. When extended, the flaps move simultaneously downward to increase the lifting force of the wing for takeoffs and landings. [Figure 3-8]

Alternate Types of Wings

Alternate types of wings are often found on aircraft. The shape and design of a wing is dependent upon the type of

operation for which an aircraft is intended and is tailored to specific types of flying. These design variations are discussed in Chapter 5, Aerodynamics of Flight, which provides information on the effect controls have on lifting surfaces from traditional wings to wings that use both flexing (due to billowing) and shifting (through the change of the aircraft's CG). For example, the wing of the weight-shift control aircraft is highly swept in an effort to reduce drag and allow for the shifting of weight to provide controlled flight. [Figure 3-9] Handbooks specific to most categories of aircraft are available for the interested pilot and can be found on the Federal Aviation Administration (FAA) website at www.faa.gov.

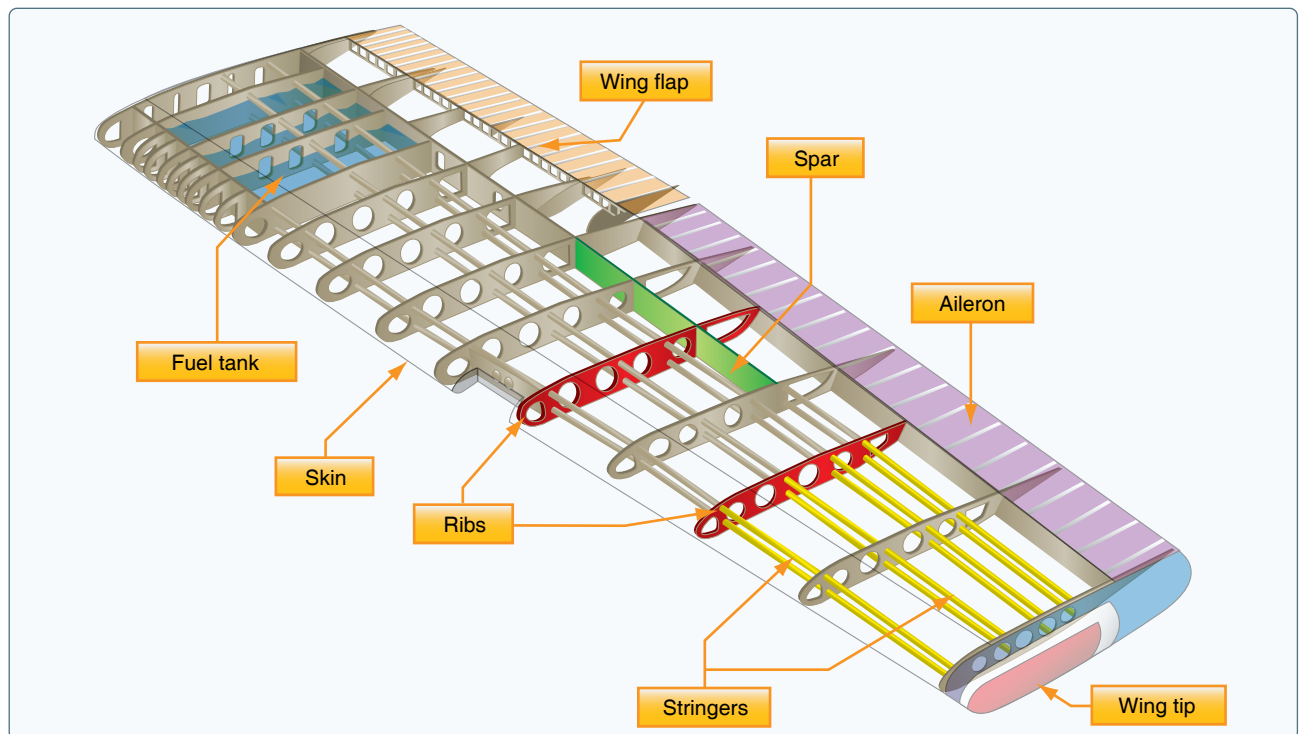


Figure 3-7. Wing components.

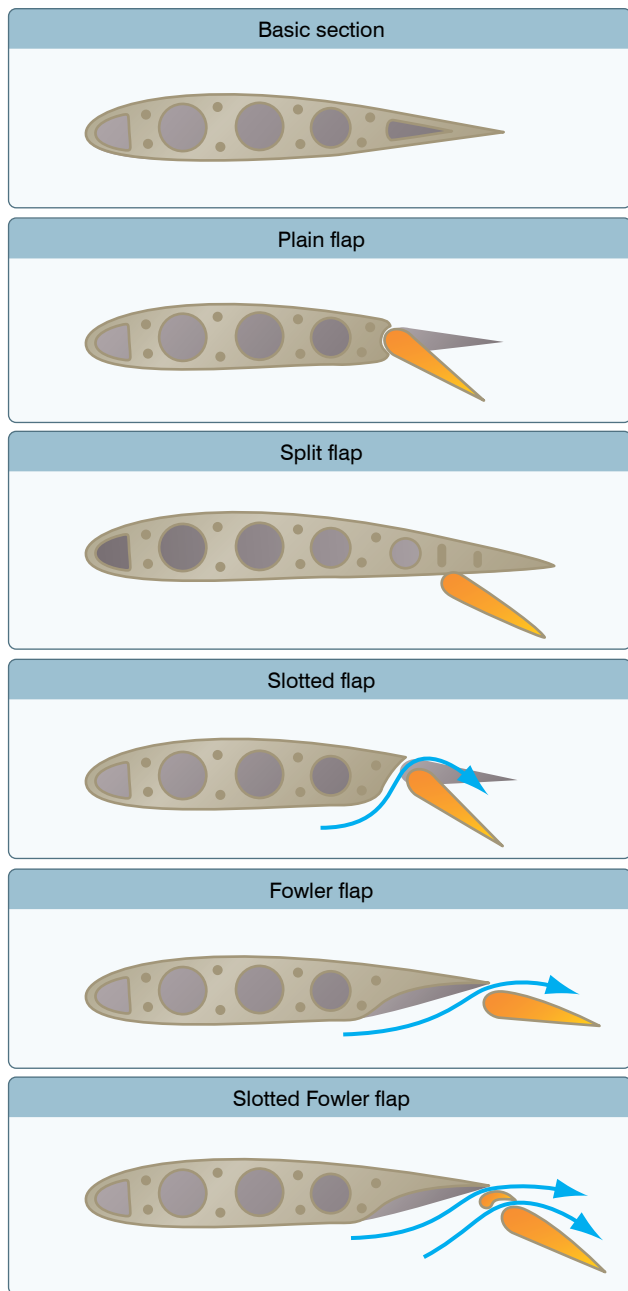


Figure 3-8. *Types of flaps.*

Empennage

The empennage includes the entire tail group and consists of fixed surfaces, such as the vertical stabilizer and the horizontal stabilizer. The movable surfaces include the rudder, the elevator, and one or more trim tabs. [Figure 3-10]

The rudder is attached to the back of the vertical stabilizer. During flight, it is used to move the airplane's nose left and right. The elevator, which is attached to the back of the horizontal stabilizer, is used to move the nose of the airplane up and down during flight. Trim tabs are small, movable



Figure 3-9. *Weight-shift control aircraft use the shifting of weight for control.*

portions of the trailing edge of the control surface. These movable trim tabs, which are controlled from the flight deck, reduce control pressures. Trim tabs may be installed on the ailerons, the rudder, and/or the elevator.

A second type of empennage design does not require an elevator. Instead, it incorporates a one-piece horizontal stabilizer that pivots from a central hinge point. This type of design is called a stabilator and is moved using the control wheel, just as the elevator is moved. For example, when a pilot pulls back on the control wheel, the stabilator pivots so the trailing edge moves up. This increases the aerodynamic tail load and causes the nose of the airplane to move up. Stabilators have an antiservo tab extending across their trailing edge. [Figure 3-11]

The antiservo tab moves in the same direction as the trailing edge of the stabilator and helps make the stabilator less sensitive. The antiservo tab also functions as a trim tab to relieve control pressures and helps maintain the stabilator in the desired position.

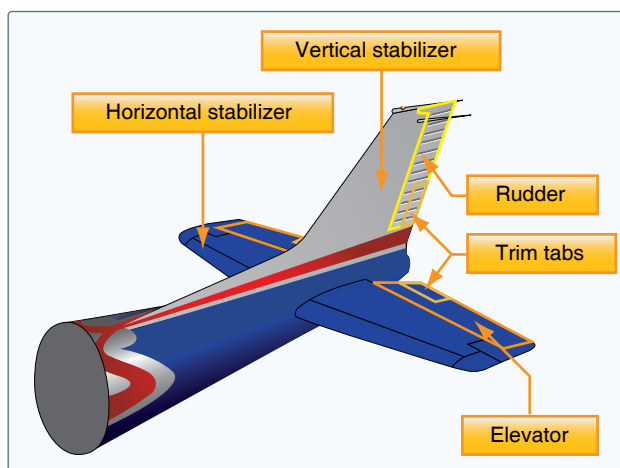


Figure 3-10. *Empennage components.*